GENERALISED KILLING SPINORS AND EUCLIDEAN HYPERSURFACES

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Abstract. We study when the number of spinors satisfying the same generalised Killing equation for a symmetric endomorphism field W forces local realisation as a Euclidean hypersurface. Let $N = 2^{\lfloor n/2 \rfloor}$ be the rank of the complex spinor bundle. We show that in dimension $n \geq 4$, if the complex dimension of the solution space exceeds $N/2$, then the metric and W satisfy the Gauss–Codazzi equations. This is proved in the smooth category by combining the curvature defects into a single bivector and bounding the dimension of its kernel on spinors, using algebraic curvature symmetries to handle exceptional cases. Examples obtained from non-flat four-dimensional hyperkähler manifolds show that this threshold is sharp for every $n \geq 4$. In dimension three, the Heisenberg group carries the maximal number of common solutions while failing the Euclidean Gauss equation. Further, in the real analytic category, we use existing extension theorems for Einstein metrics and parallel spinors to extend all common solutions to parallel spinors on a single local Ricci-flat ambient manifold. Using Wang's counts of parallel spinors, we obtain an independent proof of the sharp threshold for real analytic data.

1. Introduction

The classical hypersurface problem asks when a Riemannian manifold (M^n, g), equipped with a symmetric endomorphism field $W \in \Gamma(\mathrm{End}(TM))$, admits a local isometric immersion into $\mathbb{R}^{n+1}$ with shape operator W. The fundamental theorem of hypersurfaces gives a complete local answer to this problem: such an immersion exists if and only if g and W satisfy the Gauss–Codazzi equations. Explicitly, these can be written as follows.

$$R(X,Y)U = g(W(Y),U)W(X) - g(W(X),U)W(Y), \tag{1.1}$$

$$(\nabla_X W)(Y) = (\nabla_Y W)(X). \tag{1.2}$$

Spinors provide another way of characterising such hypersurface data. This approach was first developed by Sullivan [Sul86] for minimal surfaces in $\mathbb{R}^3$, which he described and reconstructed using spinors, and was subsequently extended by Kusner–Schmitt [KS96] to general conformal immersions. Friedrich [Fri98] gave an intrinsic characterisation. He showed that restricting a non-zero parallel spinor of the ambient space to an oriented immersed surface $M^2 \to \mathbb{R}^3$, with a suitable identification of the spinor bundles, gives a non-zero solution of constant length to the Dirac equation with potential

$$D\psi = H\psi,$$

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where H is the mean curvature function. Conversely, a non-zero solution of constant length determines a smooth symmetric endomorphism field W through

$$g(W(X),Y) = -\frac{2}{|\psi|^2}\operatorname{Re}\langle \nabla^{\Sigma M}_X \psi, Y\cdot\psi\rangle,$$

where $\langle\cdot,\cdot\rangle$ is the Hermitian spinor inner product. This field satisfies $\operatorname{tr}(W) = 2H$, and ψ satisfies the following equation, now called the generalised Killing spinor equation:

$$\nabla^{\Sigma M}_X \psi = -\frac{1}{2}W(X)\cdot\psi, \qquad \forall X \in \Gamma(TM). \tag{1.3}$$

In dimension two, the integrability conditions of this equation imply that g and W satisfy the Gauss–Codazzi equations. Hence one obtains a local isometric immersion with shape operator W.

Morel [Mor05] extended the surface correspondence to S^3 and H^3. He also showed that, in dimension three, a single non-zero generalised Killing spinor gives a local isometric immersion into $\mathbb{R}^4$ with shape operator W if W is assumed to be a Codazzi tensor. Lawn–Roth [LR10] obtained a characterisation using two spinors, without imposing the Codazzi equation separately. In the Euclidean case, their criterion requires non-zero spinors ψ_+ and ψ_-, one for W and one for $-W$, satisfying

$$\nabla^{\Sigma M}_X \psi_\pm = \mp\frac{1}{2}W(X)\cdot\psi_\pm, \qquad \forall X \in \Gamma(TM).$$

Their existence forces the Gauss–Codazzi equations and hence gives a local isometric immersion into $\mathbb{R}^4$.

In arbitrary dimension and codimension, Bayard–Lawn–Roth [BLR17] gave a spinorial representation of submanifolds in Riemannian space forms. In their Euclidean formulation, the twisted spinor bundle is constructed using left multiplication on the full real Clifford algebra. One solution is enough to recover the immersion, provided its values in a spin frame lie in the ambient spin group. Here we work with the intrinsic spinor bundle, and impose no such algebraic condition on solutions of (1.3).

A related question asks whether a generalised Killing spinor extends to a parallel spinor on an ambient manifold. Bär–Gauduchon–Moroianu [BGM05] answered this affirmatively when W is a Codazzi tensor, by constructing an explicit generalised cylinder carrying such an extension. In the Riemannian setting, the resulting ambient metric is Ricci-flat. Ammann–Moroianu–Moroianu [AMM13] removed the Codazzi assumption in the real analytic category: if g and W are real analytic, a single non-zero generalised Killing spinor extends to a parallel spinor on a local Ricci-flat ambient manifold realising (g, W) as hypersurface data. They also showed that this extension need not exist for general smooth data. In ambient dimensions six to eight, related extension results are expressed through hypo structures [CS07] and Hitchin's evolution equations for differential forms [Hit01].

For Euclidean hypersurfaces in higher dimensions, the existence of a single generalised Killing spinor in the intrinsic spinor bundle does not by itself force the Gauss–Codazzi equations. We call the differences between the two sides of (1.1) and (1.2) the Gauss and Codazzi defects, respectively. In the curvature identity associated with equation (1.3), the Gauss defect appears as a degree-two Clifford term and the Codazzi defect as a degree-one term, both acting on the same spinor. The two terms may cancel on a spinor even when neither is zero. For $n \geq 4$, we know that taking the maximal possible number of independent solutions forces the Gauss–Codazzi equations (Lemma 2.3). It is natural to ask how many solutions

we actually need. We look for the smallest number of independent solutions for one fixed W that forces these equations. Dimension three is different: Proposition C shows that even maximal multiplicity is not enough.

For a symmetric endomorphism field W, we define the vector space of its associated generalised Killing spinors by

$$\mathcal{K}_W(M) := \left\{\psi \in \Gamma(\Sigma M) : \nabla^{\Sigma M}_X \psi = -\tfrac{1}{2} W(X)\cdot\psi \quad \forall X \in \Gamma(TM)\right\}.$$

The complex rank of the spinor bundle is

$$N := \operatorname{rank}_{\mathbb{C}} \Sigma M = 2^{\lfloor n/2 \rfloor}.$$

In what follows, what matters is not the number of solutions gathered over different fields W, but the number of common solutions for a single fixed field W.

Theorem A (Sharp threshold in the smooth category)**.** *Let (M^n, g) be a smooth, connected Riemannian spin manifold of dimension $n \geq 4$, and let W be a smooth field of symmetric endomorphisms of the tangent bundle TM. If*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) > N/2,$$

then the Euclidean Gauss and Codazzi equations are satisfied. Consequently, (g, W) is locally realised as the induced metric and shape operator of an isometric immersion as a hypersurface in $\mathbb{R}^{n+1}$. Equivalently, the hypothesis is $\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq 2^{\lfloor n/2 \rfloor - 1} + 1$.

The sharpness examples use the restriction of ambient parallel spinors. The Ricci-flat extensions constructed in [AMM13] need not be flat, which suggests looking for non-flat Ricci-flat manifolds with as many parallel spinors as possible. Restricting them to suitable hypersurfaces produces many solutions for one shape operator W; we choose the hypersurfaces so that a non-zero tangential component of the ambient curvature remains as a defect in the Euclidean Gauss equation.

Theorem B (Sharpness of the threshold)**.** *In every dimension $n \geq 4$ there exist smooth data (M^n, g, W) with*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = N/2$$

which do not satisfy the Euclidean Gauss equation. These examples can also be taken real analytic.

Wang's count [Wan89] gives a simple way to see why dimension three is different. A non-flat hyperkähler four-manifold carries two independent parallel spinors in the same half-spin bundle. Restricting these to a suitably oriented three-dimensional hypersurface gives two common generalised Killing spinors, the maximal number in this dimension. We can choose the tangent space at one point to contain a plane of non-zero ambient sectional curvature, so the Euclidean Gauss equation fails. We give an explicit example on the Heisenberg group below.

Proposition C (Failure in dimension three)**.** *The three-dimensional Heisenberg group carries a left-invariant metric g and a left-invariant symmetric endomorphism field W with*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = 2 = N$$

which do not satisfy the Euclidean Gauss equation. Its generalised Killing spinors are among Artacho's examples of left-invariant spinors [Art25].

Another consequence of Theorem A is a local multiplicity gap. Let (M^n, g, W) be as in Theorem A, and let $U \subset M$ be a connected, simply connected open set. Then

$$\dim_{\mathbb{C}} \mathcal{K}_W(U) \in \{0, 1, \dots, N/2\} \cup \{N\}. \tag{1.4}$$

In the smooth proof, we adjoin a formal unit normal to the tangent space and combine the two defects into a single real bivector. Estimating the kernel of its action on spinors gives the conclusion immediately in most dimensions. In the remaining dimensions $7, 11, 15$, we use the fact that all curvature values annihilate the same subspace of spinors. This common annihilator can only be zero, the line spanned by a Kähler form, or the three-dimensional space associated with a quaternionic structure. The pair symmetry of the Gauss defect and the first Bianchi identity force the curvature values lying in the latter two to vanish as well.

For real analytic data, we apply Koiso's metric extension theorem [Koi81] and the parallel spinor extension theorem of Ammann–Moroianu–Moroianu [AMM13] to the whole solution space for fixed data (g, W). All common solutions extend to a single ambient metric, on which we can then count parallel spinors. We therefore obtain the sharp multiplicity threshold by two independent arguments: a pointwise algebraic proof for smooth data, and a proof for real analytic data using Ricci-flat extensions and holonomy.

For a Riemannian spin manifold Z, write $\mathcal{P}(Z)$ for the complex vector space of parallel spinors and, when $\dim Z$ is even, $\mathcal{P}^{\pm}(Z)$ for the subspaces of positive and negative chirality. For an oriented hypersurface $M^n \subset Z^{n+1}$ with n odd, $\tau \in \{+, -\}$ denotes the half-spin choice compatible with our intrinsic Clifford multiplication and the hypersurface identification in (3.1).

Theorem D (Analytic extension and parallel spinor counts)**.** *Let (M^n, g) be a connected real analytic Riemannian spin manifold, and let W be a real analytic symmetric endomorphism field with $\mathcal{K}_W(M) \neq 0$. On a sufficiently small connected, simply connected neighbourhood U of each point, the following hold.*

(i) *For some $\varepsilon > 0$, there is a real analytic Ricci-flat metric $\bar{g} = dt^2 + g_t$ on $Z = (-\varepsilon, \varepsilon) \times U$ whose initial slice has induced metric g and shape operator W with respect to ∂_t. This metric is uniquely determined near the initial slice by (g, W) in these Gaussian coordinates.*

(ii) *Every spinor in $\mathcal{K}_W(U)$ extends uniquely to a parallel spinor for this same ambient metric, in the appropriate spinor bundle. Restriction induces isomorphisms*

$$\mathcal{P}(Z) \cong \mathcal{K}_W(U), \quad n \text{ even},$$
$$\mathcal{P}^{\tau}(Z) \cong \mathcal{K}_W(U), \quad n \text{ odd}.$$

(iii) *If $n \geq 4$ and $\bar{g}$ is non-flat, then*

$$\dim_{\mathbb{C}} \mathcal{K}_W(U) \leq N/2.$$

For every $n \geq 4$, there are examples with $\bar{g}$ non-flat and $\dim_{\mathbb{C}} \mathcal{K}_W(U) = N/2$.

For the non-flat upper bound, we use the local de Rham decomposition and Wang's parallel-spinor counts [Wan89, MS00]. We explain how the counts on the individual factors determine the number of parallel spinors on their product.

Section 2 fixes conventions and the curvature defects, and Section 3 gives the examples. Section 4 proves the smooth threshold, and Section 5 treats the analytic extension and the count estimates. Appendix A proves the combinatorial bound on bivector kernels used in Section 4, together with the characterisation of its exceptional cases.

2. Intrinsic generalised Killing data and curvature defects

2.1. **The common equation and the modified connection.** We take the curvature and Clifford multiplication conventions

$$R(X,Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}, \qquad X \cdot Y + Y \cdot X = -2g(X,Y). \tag{2.1}$$

The spinor bundle carries the standard Hermitian inner product, and Clifford multiplication by real tangent vectors is skew-Hermitian. For an isometric hypersurface immersion $F : M \to Z$, the shape operator is defined by

$$dF(W(X)) = -\bar{\nabla}_{dF(X)}\nu,$$

where $\bar{\nabla}$ is the ambient Levi-Civita connection and ν is the chosen unit normal. Identifying the tangent space with its image, $W(X) = -\bar{\nabla}_X \nu$.

Definition 2.1. The modified connection associated with W is defined by

$$\nabla^W_X := \nabla^{\Sigma M}_X + \frac{1}{2} W(X)\,. \tag{2.2}$$

The correction term is skew-Hermitian, so this connection preserves the Hermitian inner product, and $\mathcal{K}_W(M)$ is exactly the space of ∇^W-parallel sections. Hence the lengths of solutions and their inner products are constant. On connected M, a ∇^W-parallel section that vanishes at one point vanishes identically, since parallel transport determines its value along every curve from that point. Hence at each $p \in M$ the evaluation map

$$\mathrm{ev}_p : \mathcal{K}_W(M) \longrightarrow \Sigma_p M, \qquad \psi \longmapsto \psi(p) \tag{2.3}$$

is injective, and $\dim_{\mathbb{C}} \mathcal{K}_W(M) \leq N = \mathrm{rank}_{\mathbb{C}}\, \Sigma M$. In particular, common solutions that are globally linearly independent are linearly independent at every point.

2.2. **The Gauss and Codazzi defects.** We define the following tensors.

$$(W \wedge W)(X,Y)U := g(W(Y),U)W(X) - g(W(X),U)W(Y), \tag{2.4}$$

$$\mathcal{G}_W(X,Y) := R(X,Y) - (W \wedge W)(X,Y), \tag{2.5}$$

$$\mathcal{C}_W(X,Y) := (\nabla_X W)(Y) - (\nabla_Y W)(X), \tag{2.6}$$

$$\mathcal{G}_W(X,Y,U,V) := g(\mathcal{G}_W(X,Y)U, V). \tag{2.7}$$

For fixed X, Y, $\mathcal{G}_W(X,Y)$ is a skew endomorphism, whereas $\mathcal{C}_W(X,Y)$ is a tangent vector. By the fundamental theorem of hypersurfaces, $\mathcal{G}_W = \mathcal{C}_W = 0$ is necessary and sufficient for (g, W) to be locally realised as Euclidean hypersurface data. For a proof using generalised cylinders, see [BGM05].

Proposition 2.2. *The curvature of the modified connection is*

$$R^{\nabla^W}_{X,Y} = R^{\Sigma M}(X,Y) + \frac{1}{2}\mathcal{C}_W(X,Y)\cdot + \frac{1}{4}\big[W(X)\cdot, W(Y)\cdot\big]. \tag{2.8}$$

Proof. By compatibility with Clifford multiplication, for every spinor section ψ,

$$\nabla^{\Sigma M}_X(W(Y)\cdot\psi) = \nabla_X(W(Y))\cdot\psi + W(Y)\cdot\nabla^{\Sigma M}_X\psi.$$

Expand $R^{\nabla^W}_{X,Y}\psi$ using (2.2). The mixed terms containing first derivatives of ψ cancel when we subtract the expression with X and Y exchanged. The derivatives of W, together with the correction from $\nabla^W_{[X,Y]}$, give

$$\frac{1}{2}\big(\nabla_X(W(Y)) - \nabla_Y(W(X)) - W([X,Y])\big) = \frac{1}{2}\mathcal{C}_W(X,Y),$$

using the torsion-free identity $[X,Y] = \nabla_X Y - \nabla_Y X$. The terms quadratic in W give the commutator in (2.8). □

With respect to an orthonormal frame $\{e_i\}$, the spin curvature is

$$R^{\Sigma M}(X,Y)\psi = \frac{1}{4}\sum_{i,j} g(R(X,Y)e_i, e_j)\, e_i \cdot e_j \cdot \psi.$$

On the other hand, expanding $W(X) = \sum_i g(W(X), e_i)e_i$ and similarly for $W(Y)$ gives

$$\frac{1}{4}\sum_{i,j} g((W \wedge W)(X,Y)e_i, e_j)\, e_i \cdot e_j \cdot \psi = \frac{1}{4}\big(W(Y)\cdot W(X) - W(X)\cdot W(Y)\big)\cdot\psi.$$

For any spinor field ψ, the preceding identities and antisymmetry in i, j give

$$R^{\nabla^W}_{X,Y}\psi = \frac{1}{2}\sum_{i<j} \mathcal{G}_W(X,Y,e_i,e_j)\, e_i\cdot e_j\cdot\psi + \frac{1}{2}\mathcal{C}_W(X,Y)\cdot\psi. \tag{2.9}$$

If $\psi \in \mathcal{K}_W(M)$, then ψ is parallel for ∇^W, so $R^{\nabla^W}_{X,Y}\psi = 0$.

2.3. **Adjoining a formal normal.** We now turn the degree-one Codazzi term into a bivector by adjoining one extra direction. Fix a point p and set $E = T_pM$ and $\widehat{E} = E \oplus \mathbb{R}e_0$, where e_0 is a formal unit vector orthogonal to E. The entire curvature will then act through $\Lambda^2\widehat{E}$. We identify bivectors with degree-two Clifford elements by $u \wedge v = \frac{1}{2}(u\cdot v - v\cdot u)$. The map $X \mapsto X\cdot e_0$ satisfies

$$(X\cdot e_0)(Y\cdot e_0) + (Y\cdot e_0)(X\cdot e_0) = -2\langle X, Y\rangle,$$

and so extends to an isomorphism of complex Clifford algebras

$$\alpha : \mathrm{Cl}_{\mathbb{C}}(E) \longrightarrow \mathrm{Cl}^0_{\mathbb{C}}(\widehat{E}), \qquad \alpha(X) = X\cdot e_0. \tag{2.10}$$

Indeed, the image generates the even subalgebra, and both algebras have complex dimension 2^n. Moreover, $(X\cdot e_0)(Y\cdot e_0) = X\cdot Y$, so α is the identity on $\Lambda^2 E$, where we realise $\Lambda^2 E$ as a subspace of both Clifford algebras by the above identification.

After choosing a spin frame at p, identify Σ_pM with Δ_n. For even n, $\mathrm{Cl}_{\mathbb{C}}(E)$ has a unique irreducible module up to isomorphism, realised on Δ_{n+1} through α. For odd n, the two half-spin modules realise its two inequivalent irreducible representations; let τ denote the unique choice isomorphic to the chosen intrinsic Clifford module. There is therefore an isomorphism

$$\begin{aligned} &\iota : \Delta_n \longrightarrow \Delta_{n+1}, \quad n \text{ even}, \\ &\iota : \Delta_n \longrightarrow \Delta^{\tau}_{n+1}, \quad n \text{ odd}, \end{aligned} \tag{2.11}$$

satisfying

$$\iota(a\cdot\xi) = \alpha(a)\cdot\iota(\xi).$$

In either case the target has dimension N. Set

$$\widehat{V}_p := \{\iota(\psi(p)) : \psi \in \mathcal{K}_W(M)\}.$$

By (2.3), $\dim_{\mathbb{C}} \widehat{V}_p = \dim_{\mathbb{C}} \mathcal{K}_W(M)$. Under α, the degree-two term in (2.9) is unchanged, while $\mathcal{C}_W(X,Y)$ is sent to $\mathcal{C}_W(X,Y)\cdot e_0 = \mathcal{C}_W(X,Y)\wedge e_0$. Define the real bivector

$$\Omega(X,Y) := \sum_{i<j} \mathcal{G}_W(X,Y,e_i,e_j)\, e_i\wedge e_j + \mathcal{C}_W(X,Y)\wedge e_0. \tag{2.12}$$

Then, for every $\xi \in \Sigma_p M$,

$$\iota\big(R^{\nabla^W}_{X,Y}\xi\big) = \frac{1}{2}\Omega(X,Y)\cdot \iota(\xi).$$

In particular,

$$\Omega(X,Y)\cdot \widehat{V}_p = 0. \tag{2.13}$$

The induced inner product on bivectors is characterised by

$$\langle u\wedge v, U\wedge V\rangle = \langle u,U\rangle\langle v,V\rangle - \langle u,V\rangle\langle v,U\rangle.$$

Thus, for $U,V\in E$,

$$\begin{aligned} \langle \Omega(X,Y), U\wedge V\rangle &= \mathcal{G}_W(X,Y,U,V),\\ \langle \Omega(X,Y), U\wedge e_0\rangle &= g(\mathcal{C}_W(X,Y),U). \end{aligned} \tag{2.14}$$

These component identities show that $\Omega(X,Y)=0$ for all $X,Y\in E$ if and only if $\mathcal{G}_W = \mathcal{C}_W = 0$ at p. This construction is an algebraic operation at a single point, and does not assume the existence of an ambient manifold or of an actual immersion.

We have reduced the Gauss–Codazzi equations to showing that $\Omega(X,Y)=0$ for all $X,Y\in E$. Each of these bivectors annihilates the values of all common solutions. We first consider the maximal possible multiplicity, when these values span the whole spinor fibre, and show that this is sufficient to force both defects to vanish.

Lemma 2.3 (Maximal multiplicity)**.** *Let (M^n,g) be a connected smooth Riemannian spin manifold of dimension $n\geq 4$, and let W be a smooth symmetric endomorphism field. If*

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = N,$$

then g and W satisfy the Euclidean Gauss–Codazzi equations.

Proof. Fix $p\in M$. Injectivity of evaluation implies that the values of the common solutions span $\Sigma_p M$. Using (2.11), let

$$\mathrm{cl} : \Lambda^2\widehat{E} \longrightarrow \mathrm{End}_{\mathbb{C}}(\Sigma_p M)$$

denote Clifford action transported to the intrinsic spinor fibre. By (2.13), every curvature bivector annihilates all solution values, so

$$\mathrm{cl}(\Omega(X,Y)) = 0 \qquad \text{for all } X,Y\in E.$$

We need to check that only the zero bivector acts as zero on the whole module; that is, that cl is faithful. Inside $\mathrm{Cl}^0(\widehat{E})$, the real subspace $\Lambda^2\widehat{E}$ with the commutator bracket is $\mathfrak{spin}(n+1)$. Clifford multiplication gives a representation of this Lie algebra, so $\ker\mathrm{cl}$ is an ideal. For $n\geq 4$, $\mathfrak{spin}(n+1)\cong\mathfrak{so}(n+1)$ is simple. Moreover, cl is non-zero: for orthonormal $e_1,e_2\in E$,

$$\mathrm{cl}(e_1\wedge e_2)^2 = -\mathrm{id}.$$

Consequently, $\ker\mathrm{cl}=0$, and hence $\Omega(X,Y)=0$ for all $X,Y\in E$. Applying (2.14) at each point gives $\mathcal{G}_W=\mathcal{C}_W=0$ throughout M, which is precisely the Gauss–Codazzi condition. □

For $n = 3$, one of the two simple factors of $\mathfrak{spin}(4)$ acts as zero on the chosen half-spin module. Thus cl is not faithful. The Heisenberg example in Section 3 has maximal multiplicity but fails the Gauss–Codazzi equations.

3. Sharpness and the failure in dimension three

We obtain generalised Killing spinors by restricting ambient parallel spinors to a hypersurface. Let $M^n \subset Z^{n+1}$ be an oriented hypersurface of a Riemannian spin manifold, with unit normal ν and the induced spin structure. At each point of M, the normal ν replaces the formal vector e_0. The module identification (2.11) then gives a bundle isomorphism

$$\iota : \Sigma M \longrightarrow \begin{cases} (\Sigma Z)|_M, & n \text{ even}, \\ (\Sigma^\tau Z)|_M, & n \text{ odd}, \end{cases}$$

with the same half-spin choice τ. It satisfies

$$\iota(X \cdot_M \psi) = X \cdot_Z \nu \cdot_Z \iota(\psi).$$

For an ambient spinor Ψ in the indicated bundle, we define its intrinsic restriction by

$$\psi := \iota^{-1}(\Psi|_M).$$

Under the same identification, the spinorial Gauss formula [BGM05] reads

$$\big(\bar\nabla^{\Sigma Z}_X \Psi\big)\big|_M = \nabla^{\Sigma M}_X \psi + \frac{1}{2} W(X) \cdot_M \psi. \tag{3.1}$$

If Ψ is parallel, the left-hand side vanishes, so ψ satisfies (1.3).

For odd n, Clifford multiplication by ν gives the identification with the opposite half-spin bundle:

$$\begin{aligned} \iota^{\mathrm{opp}}(\psi) &:= \nu \cdot_Z \iota(\psi) \in (\Sigma^{-\tau} Z)|_M, \\ \iota^{\mathrm{opp}}(X \cdot_M \psi) &= -X \cdot_Z \nu \cdot_Z \iota^{\mathrm{opp}}(\psi). \end{aligned} \tag{3.2}$$

Under this identification, the spinorial Gauss formula has the opposite sign in its W term, since $\bar\nabla_X \nu = -W(X)$ and $\nu \cdot_Z W(X) \cdot_Z \nu = W(X)$. Thus an ambient parallel spinor in $\Sigma^{-\tau} Z$ restricts to a solution of

$$\nabla^{\Sigma M}_X \psi = +\frac{1}{2} W(X) \cdot_M \psi,$$

the generalised Killing equation for $-W$. When counting solutions for the prescribed field W, we therefore use the parallel spinors in $\Sigma^\tau Z$.

Restriction is injective on the space of ambient parallel spinors. Indeed, a parallel spinor whose restriction is zero vanishes at a point of M, and hence on the whole connected ambient neighbourhood.

3.1. **Half-rank examples in dimension four and higher.** Let X^4 be a non-flat hyperkähler four-manifold with holonomy Sp(1). For a compact example, fix a Kähler class on a K3 surface and take the unique Ricci-flat Kähler metric in that class, given by Yau's theorem [Yau78]. With a suitable orientation, X has two complex linearly independent parallel spinors, both in the same half-spinor bundle [Wan89]. That is,

$$\dim_{\mathbb{C}} \mathcal{P}^+(X) = 2, \qquad \dim_{\mathbb{C}} \mathcal{P}^-(X) = 0.$$

For $n \geq 5$ consider the product hypersurface

$$M^n = X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3} = Z^{n+1}. \tag{3.3}$$

The spin structure is induced from the ambient product. Choosing the outward unit normal in the sphere direction,

$$W|_{TX} = 0, \qquad W|_{TS^{n-4}(r)} = -r^{-1}\,\mathrm{id}. \tag{3.4}$$

Indeed, the normal is constant in the X-directions, and $\bar{\nabla}_Y \nu = r^{-1}Y$ in the sphere directions. Since $\mathcal{P}^-(X) = 0$, the product rules reduce to

$$\mathcal{P}(Z) \cong \mathcal{P}^+(X) \otimes \mathcal{P}(\mathbb{R}^{n-3}), \quad n \text{ even},$$
$$\mathcal{P}^\pm(Z) \cong \mathcal{P}^+(X) \otimes \mathcal{P}^\pm(\mathbb{R}^{n-3}), \quad n \text{ odd}.$$

When n is even, the flat factor has odd dimension $n-3$, so the number of ambient parallel spinors is

$$2 \cdot 2^{(n-4)/2} = 2^{n/2-1} = N/2.$$

When n is odd, $n-3 \geq 2$ is even and each half-spin module of the flat factor has dimension $2^{(n-5)/2}$. Hence the number of parallel spinors in each ambient half-spinor bundle is

$$2 \cdot 2^{(n-5)/2} = 2^{(n-3)/2} = N/2.$$

In either case, a suitable restriction gives $\dim_{\mathbb{C}} \mathcal{K}_W(M) \geq N/2$.

On the other hand, for $u, v, w \in TX$, (3.4) gives

$$\mathcal{G}_W(u,v)w = R^X(u,v)w.$$

Since X is non-flat, the Gauss defect does not vanish identically. In the four-dimensional case it suffices to take

$$M^4 = X^4 \times \{0\} \subset X^4 \times \mathbb{R}, \qquad W = 0. \tag{3.5}$$

The two parallel spinors give common solutions, and again $\mathcal{G}_W = R^X \neq 0$. To see that sharpness persists when the generalised Killing spinors are not parallel (equivalently, when W is not identically zero), choose $p \in X$ with $R^X_p \neq 0$ and replace the slice by the graph of a function f with $df_p = 0$ and non-degenerate Hessian at p. Its tangent space at p is T_pX, so the hypersurface Gauss equation gives $(\mathcal{G}_W)_p = R^X_p \neq 0$, while its shape operator at p is the Hessian of f, viewed as an endomorphism, for the upward normal. After shrinking around p, W remains non-degenerate. The same two ambient parallel spinors still restrict to independent solutions.

Theorem A, which is proved independently later, shows that the number of common solutions in these examples cannot exceed $N/2$. Thus each example has exactly $N/2$ common solutions, proving Theorem B. The same count holds on sufficiently small neighbourhoods of a point where the Gauss defect is non-zero, so these also give local sharpness examples. Ricci-flat metrics are real analytic in suitable coordinates [DK81, Theorem 5.2]; the product, sphere and graph constructions can all be taken real analytic.

3.2. **The Heisenberg group example.** Let $a > 0$. Use the Heisenberg group law

$$(u,v,w) * (x,y,z) = (u+x, v+y, w+z+auy)$$

on $\mathbb{R}^3$, and set

$$g = dx^2 + dy^2 + (dz - ax\,dy)^2, \qquad e_1 = \partial_x, \quad e_2 = \partial_y + ax\,\partial_z, \quad e_3 = \partial_z.$$

Left translation preserves the coframe $dx, dy, dz - ax\,dy$. Consequently, both this metric and its dual orthonormal frame $\{e_1, e_2, e_3\}$ are left-invariant. The frame satisfies

$$[e_1, e_2] = ae_3, \qquad [e_1, e_3] = [e_2, e_3] = 0.$$

By the Koszul formula, $\nabla_{e_i} e_i = 0$ for $i = 1, 2, 3$, and the non-zero connection coefficients are

$$\nabla_{e_1} e_2 = \frac{a}{2} e_3, \qquad \nabla_{e_1} e_3 = -\frac{a}{2} e_2, \qquad \nabla_{e_2} e_3 = \frac{a}{2} e_1,$$
$$\nabla_{e_2} e_1 = -\frac{a}{2} e_3, \qquad \nabla_{e_3} e_1 = -\frac{a}{2} e_2, \qquad \nabla_{e_3} e_2 = \frac{a}{2} e_1.$$

Writing $\omega_{ij}(X) = g(\nabla_X e_i, e_j)$, the spin connection in the lifted frame is

$$\nabla^{\Sigma M}_X \psi = X(\psi) + \frac{1}{4} \sum_{i,j} \omega_{ij}(X) e_i \cdot e_j \cdot \psi,$$

where $X(\psi)$ differentiates the spinor components. We take the convention $e_1 \cdot e_2 \cdot e_3 = +\mathrm{id}$, which converts the bivector actions below into vector actions. For any spinor ψ with constant components in this frame, $X(\psi) = 0$, and so

$$\begin{aligned} \nabla^{\Sigma M}_{e_1} \psi &= \frac{a}{4} e_2 \cdot e_3 \cdot \psi = -\frac{a}{4} e_1 \cdot \psi, \\ \nabla^{\Sigma M}_{e_2} \psi &= -\frac{a}{4} e_1 \cdot e_3 \cdot \psi = -\frac{a}{4} e_2 \cdot \psi, \\ \nabla^{\Sigma M}_{e_3} \psi &= -\frac{a}{4} e_1 \cdot e_2 \cdot \psi = \frac{a}{4} e_3 \cdot \psi. \end{aligned} \tag{3.6}$$

Hence for

$$W = \frac{a}{2} \operatorname{diag}(1, 1, -1) \tag{3.7}$$

with respect to $\{e_1, e_2, e_3\}$, every constant spinor is a common solution. By injectivity of evaluation,

$$\dim_{\mathbb{C}} \mathcal{K}_W(M) = 2 = N.$$

This is a concrete instance of the left-invariant generalised Killing spinors treated in [Art25]. With the opposite Clifford convention, W is replaced by $-W$. Writing the sectional curvature as $K_{ij} = g(R(e_i, e_j)e_j, e_i)$, the connection coefficients above give the first row of the following table. The second row is the product of the corresponding eigenvalues of W, as required by the Gauss equation.

	K_{12}	K_{13}	K_{23}
Actual curvature	$-3a^2/4$	$a^2/4$	$a^2/4$
Required by the Euclidean Gauss equation	$a^2/4$	$-a^2/4$	$-a^2/4$

Hence $\mathcal{G}_W \neq 0$, and Proposition C follows. A further direct computation gives $\mathcal{C}_W(e_1, e_2) = a^2 e_3$. In this example the modified connection itself is flat, but the Gauss and Codazzi defects are both non-zero.

4. Pointwise rigidity and the sharp smooth threshold

In this section we prove Theorem A. Fix $p \in M$ and use E, $\widehat{E}$ and $\widehat{V}_p$ from Section 2. Throughout the proof we assume

$$\dim_{\mathbb{C}} \widehat{V}_p = \dim_{\mathbb{C}} \mathcal{K}_W(M) > N/2.$$

By (2.13), every curvature bivector $\Omega(X, Y)$ annihilates the same subspace $\widehat{V}_p$ of the N-dimensional spin module. We first bound the kernel of a non-zero bivector. This shows that

$\Omega(X,Y) = 0$ except in dimensions 7, 11 and 15. In those cases, we classify the bivectors that annihilate all of $\widehat{V}_p$, then use the curvature symmetries to show that $\Omega(X,Y)$ must still vanish.

4.1. **Kernels of bivectors and parity-restricted zero weights.**

The eigenvalues of a bivector

Set $q = \lfloor (n+1)/2 \rfloor$, so the full spinor module Δ_{n+1} of the formal ambient space $\widehat{E}$ has complex dimension 2^q. A real bivector $A \in \Lambda^2 \widehat{E}$ can be written, with respect to a suitable orthonormal basis $\{f_1, \dots, f_{n+1}\}$, as

$$A = \sum_{j=1}^{q} \mu_j f_{2j-1} \wedge f_{2j}. \tag{4.1}$$

When $n+1$ is even and the orientation of the basis is fixed, we allow the μ_j to have either sign. The Clifford operators $f_{2j-1} \cdot f_{2j}$ commute with one another and square to $-\mathrm{id}$, so they admit a basis of simultaneous eigenspinors. For a non-zero simultaneous eigenspinor ξ, write

$$(f_{2j-1} \cdot f_{2j}) \cdot \xi = i\varepsilon_j \xi, \qquad \varepsilon_j \in \{\pm 1\}.$$

Every sign choice occurs. Indeed, multiplication by f_{2j-1} takes a simultaneous eigenspinor with signs

$$(\varepsilon_1, \dots, \varepsilon_j, \dots, \varepsilon_q) \quad \text{to one with signs} \quad (\varepsilon_1, \dots, -\varepsilon_j, \dots, \varepsilon_q).$$

This follows because f_{2j-1} anticommutes with $f_{2j-1} \cdot f_{2j}$ and commutes with the other pairs. Since $f_{2j-1}^2 = -1$, a non-zero spinor stays non-zero. We can change any signs we wish in this way, so there are 2^q simultaneous eigenspaces. Their dimensions sum to $\dim_{\mathbb{C}} \Delta_{n+1} = 2^q$, and hence each has dimension one.

On the eigenspace with signs $\varepsilon \in \{\pm 1\}^q$, the action of A is multiplication by

$$i \sum_{j=1}^{q} \varepsilon_j \mu_j. \tag{4.2}$$

It follows that $\dim_{\mathbb{C}} \ker(A\cdot)$ is exactly the number of sign choices for which this sum vanishes.

Lemma 4.1 (Kernel bound on the full spin module)**.** *If $A \neq 0$, then on the full spin module Δ_{n+1},*

$$\dim_{\mathbb{C}} \ker(A\cdot) \leq \frac{1}{2} \dim_{\mathbb{C}} \Delta_{n+1} = 2^{q-1}.$$

Proof. We may assume $\mu_1 \neq 0$. Once $\varepsilon_2, \dots, \varepsilon_q$ are fixed, at most one $\varepsilon_1 \in \{\pm 1\}$ satisfies the zero-weight condition $\varepsilon_1 \mu_1 = -\sum_{j=2}^{q} \varepsilon_j \mu_j$. Hence there are at most 2^{q-1} zero weights. □

Even intrinsic dimension

For even n, the formal ambient space has odd dimension $n+1$. In this case, (2.11) identifies the intrinsic spinor fibre with the full ambient spin module Δ_{n+1}, whose complex dimension is $2^{n/2} = N$. For each fixed pair $X, Y \in E$, (2.13) shows that the kernel of Clifford multiplication by $\Omega(X,Y)$ contains $\widehat{V}_p$, so its dimension exceeds $N/2$. Lemma 4.1 therefore forces $\Omega(X,Y) = 0$ for every $X, Y \in E$.

Odd intrinsic dimension

Now suppose n is odd, so $n = 2q-1$. The formal ambient dimension is $n+1 = 2q$, but (2.11) identifies the intrinsic spinor fibre with just one half-spin module:

$$\dim_{\mathbb{C}} \Delta_{n+1} = 2^q = 2N, \qquad \dim_{\mathbb{C}} \Delta^{\tau}_{n+1} = 2^{q-1} = N, \qquad N/2 = 2^{q-2}.$$

Lemma 4.1 bounds the kernel of a non-zero bivector on the full module by N. This does not contradict the existence of more than $N/2$ independent solution values in the chosen half. We therefore need a bound on the kernel within that half-spin module.

On Δ^{τ}_{n+1}, the signs satisfy $\prod_j \varepsilon_j = \sigma$, where $\sigma \in \{\pm 1\}$ depends on τ and our convention for the volume element. Setting

$$Z^{\sigma}(\mu) := \Big\{ \varepsilon \in \{\pm 1\}^q : \prod_j \varepsilon_j = \sigma, \quad \sum_j \varepsilon_j \mu_j = 0 \Big\},$$

we have $\dim_{\mathbb{C}} \ker(A \cdot |_{\Delta^{\tau}_{n+1}}) = |Z^{\sigma}(\mu)|$.

Dimension three: recovering Lawn–Roth

Before proving the general estimate, we illustrate the weight calculation in dimension three. Here $q = 2$ and each ambient half-spin module has complex dimension two. Suppose that there is a non-zero generalised Killing spinor for each of W and $-W$. At a fixed point, the algebraic version of (3.2) is $\iota^{\mathrm{opp}}(\xi) = e_0 \cdot \iota(\xi)$. Since $\mathcal{G}_{-W} = \mathcal{G}_W$ and $\mathcal{C}_{-W} = -\mathcal{C}_W$, the opposite intertwining rule and (2.9) give

$$\iota^{\mathrm{opp}}\big(R^{\nabla^{-W}}_{X,Y}\,\xi\big) = \frac{1}{2}\Omega(X,Y) \cdot \iota^{\mathrm{opp}}(\xi).$$

Thus the values of the two solutions give non-zero vectors annihilated by the same curvature bivector $\Omega(X,Y)$, one in each half-spin module.

For fixed X, Y, write

$$\Omega(X,Y) = \mu_1 f_1 \wedge f_2 + \mu_2 f_3 \wedge f_4.$$

The sign vectors $(+,+)$ and $(-,-)$ belong to one parity class, and $(+,-)$ and $(-,+)$ to the other. Thus, up to interchanging the half-spin labels, the eigenvalues are

$$\begin{aligned} \text{on } \Delta_4^+ : &\quad i(\mu_1+\mu_2),\ -i(\mu_1+\mu_2), \\ \text{on } \Delta_4^- : &\quad i(\mu_1-\mu_2),\ -i(\mu_1-\mu_2). \end{aligned}$$

A non-zero kernel in each half therefore requires $\mu_1 + \mu_2 = 0$ and $\mu_1 - \mu_2 = 0$, so both coefficients vanish. Since X, Y were arbitrary, $\Omega = 0$, and (2.14) gives the Gauss–Codazzi equations. This recovers the Euclidean criterion of Lawn–Roth [LR10]: one non-zero solution for each sign suffices. The Heisenberg example in Section 3 shows why both signs matter: even two independent solutions for W alone do not suffice.

The parity-restricted estimate

We now return to the higher-dimensional problem, where we must bound the kernel within a single half-spin module.

Theorem 4.2 (Parity-restricted zero-weight estimate)**.** *For $q \geq 2$, $\mu \in \mathbb{R}^q \setminus \{0\}$ and $\sigma \in \{\pm 1\}$,*

$$|Z^\sigma(\mu)| \leq \begin{cases} 2^{q-2}, & q \text{ odd}, \\ \max\Big\{2^{q-2}, \binom{q}{q/2}\Big\}, & q \text{ even}. \end{cases} \tag{4.3}$$

This bound is sharp for each q and each fixed σ. Moreover, if $q \in \{4, 6, 8\}$ and $|Z^\sigma(\mu)| > 2^{q-2}$, then all coefficients are non-zero and

$$|\mu_1| = \cdots = |\mu_q|, \qquad |Z^\sigma(\mu)| = \binom{q}{q/2}, \qquad Z^{-\sigma}(\mu) = \varnothing. \tag{4.4}$$

The proof in Appendix A uses the classical antichain method [Erd45, Spe28]. We also prove the exceptional rigidity needed for the geometric argument.

The exceptional dimensions

If $\Omega(X, Y) \neq 0$, its kernel on the chosen half-spin module contains $\widehat{V}_p$ and so has dimension greater than 2^{q-2}. For odd q, Theorem 4.2 bounds that same dimension by 2^{q-2}, a contradiction. For even q, the only cases in which the central binomial coefficient exceeds 2^{q-2} are $q = 2, 4, 6, 8$. Since Theorem A assumes $n \geq 4$, only $n = 7, 11, 15$ remain. The failure at $n = 3$ was exhibited in Proposition C.

For the remaining dimensions $n = 7, 11, 15$, we use the metric to pass between two-forms, which are evaluated on pairs of vectors, and bivectors, which act on spinors by Clifford multiplication. For an orthogonal complex structure J, its Kähler form and corresponding bivector are characterised by

$$\omega_J(X, Y) := \langle JX, Y\rangle = \langle \omega_J^\sharp, X \wedge Y\rangle, \qquad \omega_J \in \Lambda^2 \widehat{E}^*, \quad \omega_J^\sharp \in \Lambda^2 \widehat{E}.$$

If all the coefficients in (4.1) have the same absolute value $\lambda > 0$, then

$$Jf_{2j-1} = \frac{\mu_j}{\lambda} f_{2j}, \qquad Jf_{2j} = -\frac{\mu_j}{\lambda} f_{2j-1}$$

defines an orthogonal complex structure: each μ_j/λ is $+1$ or -1, so $J^2 = -\mathrm{id}$ and J preserves the metric. Then $A = \lambda\omega_J^\sharp$. By the rigidity statement in Theorem 4.2, every non-zero bivector whose kernel exceeds half rank in these exceptional dimensions has this form.

4.2. **The common annihilator.** For $n = 7, 11, 15$, the kernel bound alone is not enough. Happily, there is another condition available: every $\Omega(X, Y)$ must annihilate the same space $\widehat{V}_p$ of solution values. Accordingly, let $q \in \{4, 6, 8\}$, so that $\dim \widehat{E} = 2q$ and $N = 2^{q-1}$, and retain the hypothesis $\dim_{\mathbb{C}} \widehat{V}_p > N/2 = 2^{q-2}$.

Definition 4.3. The common annihilator of $\widehat{V}_p$ is defined as

$$\mathfrak{a}_p := \{A \in \Lambda^2 \widehat{E} : A \cdot \xi = 0 \ \ \forall \xi \in \widehat{V}_p\}.$$

This real vector space contains every curvature value $\Omega(X, Y)$. If two bivectors annihilate $\widehat{V}_p$, so does their Clifford commutator.

Lemma 4.4. *The common annihilator is one of*

$$\{0\}, \qquad \mathbb{R}\omega_J^\sharp, \qquad \operatorname{span}_{\mathbb{R}}\{\omega_I^\sharp, \omega_J^\sharp, \omega_K^\sharp\}.$$

In the last case, I, J, K are orthogonal complex structures forming a quaternionic triple with $IJ = K = -JI$.

Proof. Bivectors have even degree and hence preserve the chosen half-spin module. With the Clifford commutator $[A, B]_{\mathrm{Cl}} = AB - BA$,

$$[f_i f_j, f_k]_{\mathrm{Cl}} = 2(\delta_{ik} f_j - \delta_{jk} f_i), \tag{4.5}$$

$$[f_i f_j, f_k f_l]_{\mathrm{Cl}} = 2\big(\delta_{ik} f_j \wedge f_l - \delta_{jk} f_i \wedge f_l - \delta_{il} f_j \wedge f_k + \delta_{jl} f_i \wedge f_k\big). \tag{4.6}$$

The first formula follows by moving f_k to the left using $f_a f_b + f_b f_a = -2\delta_{ab}$. The second follows from $[P, QR] = [P, Q]R + Q[P, R]$; on substituting $f_a f_b = f_a \wedge f_b - \delta_{ab}$, the scalar terms cancel. Hence $[A, B]_{\mathrm{Cl}} \in \Lambda^2 \widehat{E}$, and if $A, B \in \mathfrak{a}_p$, then

$$[A, B]_{\mathrm{Cl}} \cdot \xi = A \cdot (B \cdot \xi) - B \cdot (A \cdot \xi) = 0 \qquad (\xi \in \widehat{V}_p).$$

So $\mathfrak{a}_p$ is closed under the Clifford commutator.

We use the correspondence between bivectors and skew endomorphisms to express the preceding rigidity as a condition on their squares. To each bivector A associate the skew map T_A defined by

$$\langle T_A X, Y\rangle = \langle A, X \wedge Y\rangle. \tag{4.7}$$

This correspondence is linear and injective, and

$$T_{u\wedge v} z = \langle u, z\rangle v - \langle v, z\rangle u, \qquad T_{\omega_J^\sharp} = J.$$

From (4.5) and linearity,

$$[A, z]_{\mathrm{Cl}} = 2T_A z. \tag{4.8}$$

Applying the Jacobi identity,

$$2T_{[A,B]_{\mathrm{Cl}}} z = [[A, B]_{\mathrm{Cl}}, z]_{\mathrm{Cl}} = 4(T_A T_B - T_B T_A) z,$$

that is,

$$T_{[A,B]_{\mathrm{Cl}}} = 2[T_A, T_B]. \tag{4.9}$$

If $0 \neq A \in \mathfrak{a}_p$, its kernel contains $\widehat{V}_p$ and therefore exceeds half rank. By the rigidity in Theorem 4.2 we can write $A = \lambda\omega_J^\sharp$, so $T_A = \lambda J$ and $T_A^2 = -\lambda^2 \mathrm{id}$. Here $\dim \widehat{E} = 2q$ and $|A|^2 = q\lambda^2$. Setting

$$Q(A) := -\frac{1}{2q}\operatorname{tr}(T_A^2) = \frac{|A|^2}{q}, \tag{4.10}$$

we have, for every $A \in \mathfrak{a}_p$ including zero,

$$T_A^2 = -Q(A)\,\mathrm{id}. \tag{4.11}$$

Applying (4.11) to $A + B$ and subtracting the identities for A and B gives

$$T_A T_B + T_B T_A = -2b(A, B)\,\mathrm{id}, \qquad b(A, B) = \frac{1}{q}\langle A, B\rangle. \tag{4.12}$$

If $\dim_{\mathbb{R}} \mathfrak{a}_p \geq 2$, choose A, B orthonormal with respect to b, and set $I = T_A$ and $J = T_B$. Then $I^2 = J^2 = -\mathrm{id}$ and $IJ = -JI$, and

$$K := IJ = \frac{1}{2}[I, J] = T_{\frac{1}{4}[A,B]_{\mathrm{Cl}}}$$

also corresponds to an element of $\mathfrak{a}_p$. K is skew and satisfies $K^2 = -\mathrm{id}$, and I, J, K form a quaternionic triple. For example, $IK + KI = 0$, so by (4.12) the corresponding bivectors are mutually b-orthogonal.

If there were a further non-zero element orthogonal to this triple, its normalised associated map L would anticommute with each of I, J, K. But anticommutation with I and J also gives $LK = LIJ = IJL = KL$. Hence $KL = 0$, contradicting $K^2 = L^2 = -\mathrm{id}$. The two-dimensional case does not occur either, since two orthogonal generators produce the third, so the possible dimensions are 0, 1 and 3. Applying (4.11) to the generator in the one-dimensional case as well gives the stated classification. □

4.3. **Eliminating the exceptions by curvature symmetries.** We continue to assume $q \in \{4, 6, 8\}$ and $\dim_{\mathbb{C}} \widehat{V}_p > 2^{q-2}$. Set $h(X, Y) = g(W(X), Y)$. By (2.4) and the symmetry of W,

$$g((W \wedge W)(X, Y)U, V) = h(Y, U)h(X, V) - h(X, U)h(Y, V)$$

has pair symmetry. Moreover, in the cyclic sum over X, Y, U all terms cancel by the symmetry of h. Hence $\mathcal{G}_W$, obtained by subtracting this term from R, also satisfies

$$\mathcal{G}_W(X, Y, U, V) = \mathcal{G}_W(U, V, X, Y), \tag{4.13}$$

$$\mathcal{G}_W(X, Y, U, V) + \mathcal{G}_W(Y, U, X, V) + \mathcal{G}_W(U, X, Y, V) = 0. \tag{4.14}$$

In what follows, all curvature arguments lie in $E = e_0^{\perp}$. We use ω_J also for its restriction to $E \times E$, while $\omega_J^{\sharp}$ continues to denote the ambient bivector in $\Lambda^2 \widehat{E}$. By (2.14),

$$\mathcal{G}_W(X, Y, U, V) = \langle \Omega(X, Y), U \wedge V \rangle.$$

Lemma 4.5 (The Kähler line case)**.** *If $\mathfrak{a}_p = \mathbb{R}\omega_J^{\sharp}$, then $\Omega(X, Y) = 0$ for all $X, Y \in E$.*

Proof. Since Ω is alternating and bilinear in X, Y and takes values in the fixed line $\mathbb{R}\omega_J^{\sharp}$, its coefficient is a two-form $\beta \in \Lambda^2 E^*$. Thus $\Omega(X, Y) = \beta(X, Y)\omega_J^{\sharp}$, and the preceding identity gives

$$\mathcal{G}_W(X, Y, U, V) = \beta(X, Y)\omega_J(U, V).$$

The subspace $H = \mathrm{span}\{e_0, Je_0\}$ and its orthogonal complement are J-invariant. Choosing a unit vector $u \in H^{\perp}$, the vectors u and Ju lie in E and satisfy $\omega_J(u, Ju) = \langle Ju, Ju \rangle = 1$. Setting $(U, V) = (u, Ju)$ in the pair symmetry gives, for every $X, Y \in E$,

$$\beta(X, Y) = \beta(u, Ju)\omega_J(X, Y).$$

Hence $\beta = c\omega_J$ on E, where $c = \beta(u, Ju)$, and $\mathcal{G}_W = c\,\omega_J \otimes \omega_J$.

The space $\mathrm{span}\{e_0, Je_0, u, Ju\}$ is J-invariant and has dimension four. Its orthogonal complement is therefore J-invariant and has dimension $2q - 4 \geq 4$. Choose a unit vector v there. Then u, Ju, v, Jv are orthonormal tangent vectors. Applying (4.14) to this quadruple,

$$0 = c\big(\omega_J(u, Ju)\omega_J(v, Jv) + \omega_J(Ju, v)\omega_J(u, Jv) + \omega_J(v, u)\omega_J(Ju, Jv)\big) = c.$$

The last two terms vanish by the orthogonality of distinct J-invariant planes. Hence $c = 0$, $\beta = 0$ and $\Omega = 0$. □

Lemma 4.6 (The quaternionic triple case)**.** *If* $\mathfrak{a}_p = \operatorname{span}_{\mathbb{R}}\{\omega_I^\sharp, \omega_J^\sharp, \omega_K^\sharp\}$*, then* $\Omega(X,Y) = 0$ *for all* $X, Y \in E$*.*

Proof. Set $(I_1, I_2, I_3) = (I, J, K)$ and $\omega_a = \omega_{I_a}$, using the same restriction convention as above. For some two-forms $\beta_a \in \Lambda^2 E^*$ we can write

$$\Omega(X,Y) = \sum_a \beta_a(X,Y)\omega_a^\sharp, \qquad \mathcal{G}_W(X,Y,U,V) = \sum_a \beta_a(X,Y)\omega_a(U,V).$$

The quaternionic four-dimensional subspace

$$H_0 := \operatorname{span}\{e_0, Ie_0, Je_0, Ke_0\}$$

and its orthogonal complement are invariant under each I_a. Here the dimension hypothesis provides room for a second quaternionic four-plane: since $\dim \widehat{E} \geq 8$, choose a unit vector $v \in H_0^\perp$. Then $H_1 = \operatorname{span}\{v, Iv, Jv, Kv\}$ is also a quaternionic four-dimensional subspace, with $H_1 \subset E$ and $H_1 \perp H_0$. The quaternionic relations and orthogonality give

$$\omega_a(v, I_b v) = \langle I_a v, I_b v\rangle = \delta_{ab}. \tag{4.15}$$

In particular, the restrictions of $\omega_1, \omega_2, \omega_3$ to E are linearly independent.

Setting $(U, V) = (v, I_a v)$ in the pair symmetry gives

$$\beta_a(X,Y) = \sum_b \beta_b(v, I_a v)\omega_b(X,Y).$$

Defining $S_{ab} := \beta_b(v, I_a v)$, we get $\beta_a = \sum_b S_{ab}\omega_b$. Evaluating this on $(v, I_b v)$ and comparing with the definition gives $S_{ab} = S_{ba}$. Hence

$$\mathcal{G}_W(X,Y,U,V) = \sum_{r,s} S_{rs}\,\omega_s(X,Y)\omega_r(U,V). \tag{4.16}$$

If $x \in H_0 \cap E$ and $y \in H_1$, then $\omega_s(x,y) = \langle I_s x, y\rangle = 0$, since $I_s x \in H_0$. Equation (4.16) therefore gives $\mathcal{G}_W(x,y,U,V) = 0$ whenever $x \in H_0 \cap E$, $y \in H_1$ and $U, V \in E$. To show that every S_{ab} vanishes, choose $a, b \in \{1,2,3\}$ independently, and choose c, d so that (b,c,d) is one of $(1,2,3)$, $(2,3,1)$ or $(3,1,2)$. By the quaternionic relations and (4.15),

$$\omega_s(I_c e_0, I_d e_0) = \delta_{sb}, \qquad \omega_r(v, I_a v) = \delta_{ra}.$$

In (4.16), the pair $(I_c e_0, I_d e_0)$ selects column b, and the pair $(v, I_a v)$ selects row a. Only the term S_{ab} remains. All four vectors lie in E. From (4.14) and (4.16),

$$\begin{aligned} 0 &= \mathcal{G}_W(I_c e_0, I_d e_0, v, I_a v) \\ &\quad + \mathcal{G}_W(I_d e_0, v, I_c e_0, I_a v) + \mathcal{G}_W(v, I_c e_0, I_d e_0, I_a v) \\ &= S_{ab}. \end{aligned}$$

The last two terms vanish because their first two inputs lie in $H_0 \cap E$ and H_1, respectively, or in the reverse order. Since a and b can be chosen independently, every S_{ab} vanishes. Hence $\beta_a = 0$ and $\Omega = 0$. □

Proof of Theorem A. At every $p \in M$, $\dim_{\mathbb{C}} \widehat{V}_p = \dim_{\mathbb{C}} \mathcal{K}_W(M) > N/2$. If n is even, Lemma 4.1 gives $\Omega(X,Y) = 0$. If n is odd and $n \notin \{3, 7, 11, 15\}$, Theorem 4.2 gives the same conclusion. For the remaining cases $n = 7, 11, 15$ we use Lemma 4.4. If the annihilator is zero, the conclusion is immediate; if it is a line or a quaternionic triple, we apply Lemmas 4.5 and 4.6. Hence $\Omega = 0$ in all cases. The component identities (2.14) give $\mathcal{G}_W = \mathcal{C}_W = 0$. By the fundamental theorem of hypersurfaces, the required local isometric immersion exists. □

Corollary 4.7 (Local multiplicity gap)**.** *Let (M^n, g) be a connected smooth Riemannian spin manifold, $n \geq 4$, and let W be a smooth symmetric endomorphism field of TM. For every connected, simply connected open set $U \subset M$, the solution space for $W|_U$ satisfies*

$$\dim_{\mathbb{C}} \mathcal{K}_W(U) \in \{0, 1, \ldots, N/2\} \cup \{N\}.$$

Proof. If $\dim_{\mathbb{C}} \mathcal{K}_W(U) > N/2$, Theorem A gives $\mathcal{G}_W = \mathcal{C}_W = 0$, so (2.9) gives $R^{\nabla^W} = 0$. Fix $p \in U$ and $\xi \in \Sigma_p M$. For $x \in U$, define $\psi(x)$ by transporting ξ from p to x. Any two paths from p to x are homotopic with fixed endpoints, since U is simply connected. Flatness then implies that they give the same parallel transport. This defines the unique ∇^W-parallel section with $\psi(p) = \xi$. Thus evaluation is an isomorphism $\mathcal{K}_W(U) \cong \Sigma_p M$, and $\dim_{\mathbb{C}} \mathcal{K}_W(U) = N$. □

Without simple connectedness, the hypothesis $\dim_{\mathbb{C}} \mathcal{K}_W(U) > N/2$ still gives $R^{\nabla^W} = 0$, but parallel transport around a loop need not be the identity. Fix $p \in U$. The monodromy representation $\rho_p : \pi_1(U, p) \to \mathrm{GL}(\Sigma_p M)$ assigns to each homotopy class the ∇^W-parallel transport around a representative loop based at p. A spinor value at p extends to a global parallel section if and only if it returns to itself around every loop. Evaluation therefore identifies

$$\mathcal{K}_W(U) \cong \{\xi \in \Sigma_p M : \rho_p([\gamma])\xi = \xi \text{ for every } [\gamma] \in \pi_1(U, p)\}.$$

If the monodromy is non-trivial, this fixed space is a proper subspace of $\Sigma_p M$. It follows that $\dim_{\mathbb{C}} \mathcal{K}_W(U) < N$. This is why simple connectedness is needed for the conclusion above.

5. Analytic extension and parallel spinor counts

Multiple common solutions constrain the holonomy of an ambient extension only if they extend to the same metric. For real analytic data, the metric extension is determined by (g, W), and the extension theorem of Ammann–Moroianu–Moroianu [AMM13] applies to every common solution on this same metric. We can therefore use the holonomy classification to bound the number of solutions when the ambient metric is non-flat, keeping track of chirality in even ambient dimension. This proves Theorem D and gives an independent proof of the sharp threshold for real analytic data.

5.1. **The common Ricci-flat germ.** We first construct the ambient metric. Given a connected Riemannian spin manifold (M^n, g) and a symmetric endomorphism field W, we want a Ricci-flat metric $\bar{g}$ in one higher dimension such that M has induced metric g and shape operator W. Locally, take $Z = (-\varepsilon, \varepsilon) \times U$, where $U \subset M$ is a sufficiently small contractible neighbourhood. Identifying U with the initial slice $\{0\} \times U$, we use Gaussian coordinates:

$$\bar{g} = dt^2 + g_t, \qquad g_0 = g|_U, \qquad \dot{g}_0(X, Y) = -2g(W(X), Y). \tag{5.1}$$

Here ∂_t is the unit normal to the slices, and $t \mapsto (t, x)$ is a unit-speed normal geodesic. The last identity ensures that the initial slice has shape operator W, with our convention $W(X) = -\bar{\nabla}_X \partial_t$. If the ambient metric is Ricci-flat, contracting the Gauss and Codazzi equations gives the constraints (5.2) below on (g, W). Write ∇^g for the Levi-Civita connection and define the codifferential of W by

$$(\delta W)(X) := -\sum_i g((\nabla^g_{e_i} W)(e_i), X)$$

for a local orthonormal frame $\{e_i\}$. A single non-zero generalised Killing spinor supplies these constraints; see the proof of [AMM13, Corollary 3.1]. We recall the calculation using the Schrödinger–Lichnerowicz formula.

Proposition 5.1. *If $\mathcal{K}_W(M) \neq 0$, then*

$$d\operatorname{tr}(W) + \delta W = 0, \qquad \operatorname{Scal}_g + \operatorname{tr}(W^2) - (\operatorname{tr}(W))^2 = 0. \tag{5.2}$$

Proof. Take $0 \neq \psi \in \mathcal{K}_W(M)$. Since M is connected, ψ has non-zero constant length. Write ∇ for the spin connection in this proof. Symmetry of W, Clifford contraction and the Leibniz rule give

$$D\psi = \frac{1}{2}\operatorname{tr}(W)\psi, \qquad D^2\psi = \frac{1}{2}(d\operatorname{tr}(W))^\sharp \cdot \psi + \frac{1}{4}(\operatorname{tr}(W))^2\psi.$$

Recall that the connection Laplacian is

$$\nabla^*\nabla\psi = -\sum_i \left(\nabla_{e_i}\nabla_{e_i}\psi - \nabla_{\nabla^g_{e_i}e_i}\psi\right).$$

Choosing a local orthonormal frame that is geodesic at a point, and differentiating the generalised Killing equation there, gives

$$\begin{aligned}\nabla^*\nabla\psi &= \frac{1}{2}\sum_i (\nabla^g_{e_i}W)(e_i)\cdot\psi - \frac{1}{4}\sum_i W(e_i)\cdot W(e_i)\cdot\psi \\ &= -\frac{1}{2}(\delta W)^\sharp\cdot\psi + \frac{1}{4}\operatorname{tr}(W^2)\psi.\end{aligned}$$

Applying the Schrödinger–Lichnerowicz formula $D^2 = \nabla^*\nabla + \operatorname{Scal}_g/4$ [Lic63] yields

$$\left(\operatorname{Scal}_g + \operatorname{tr}(W^2) - (\operatorname{tr}(W))^2\right)\psi = 2(d\operatorname{tr}(W) + \delta W)^\sharp\cdot\psi. \tag{5.3}$$

Taking the real part of the Hermitian inner product with ψ shows that the scalar coefficient on the left vanishes, since Clifford multiplication by real vectors is skew-Hermitian and $|\psi|^2 > 0$. Writing $V = (d\operatorname{tr}(W) + \delta W)^\sharp$, the remaining equation is $V\cdot\psi = 0$. Since $V\cdot V = -|V|^2$, we obtain $V = 0$. This proves (5.2). □

Assume now that (g, W) is real analytic and $\mathcal{K}_W(M) \neq 0$. Koiso's Einstein extension theorem [Koi81], in the form stated in [AMM13, Theorem 2.1] with Einstein constant zero, provides a unique real analytic Ricci-flat metric germ with initial data (5.1). In Gaussian coordinates, the tangential Ricci-flat equations determine $\ddot{g}_t$ in terms of g_t, $\dot{g}_t$ and tangential derivatives. With g_0 and $\dot{g}_0$ as initial data, the Cauchy–Kowalevski theorem gives a unique solution g_t for sufficiently small $|t|$. The constraints (5.2) are preserved along the solution, so

$$\operatorname{Ric}_{\bar{g}}(\partial_t, \cdot) = 0.$$

Together with the tangential equations, this gives $\operatorname{Ric}_{\bar{g}} = 0$. In fixed Gaussian coordinates, two metrics define the same *germ* if they agree on a smaller neighbourhood of the initial slice. Intrinsically, any two real analytic Ricci-flat extensions $(Z_1, \bar{g}_1)$ and $(Z_2, \bar{g}_2)$ with the same initial data are related, after shrinking around U, by a diffeomorphism $\Phi : Z_1 \to Z_2$ satisfying

$$\Phi^*\bar{g}_2 = \bar{g}_1, \qquad \Phi|_U = \mathrm{id}_U, \qquad d\Phi(\nu_1) = \nu_2,$$

where ν_i is the chosen unit normal along the initial slice in Z_i. We may thus work with a metric $\bar{g}$ on $Z = (-\varepsilon, \varepsilon)\times U$, shrinking U and ε as needed. Its construction uses only g and W, so we keep $\bar{g}$ fixed as ψ varies in $\mathcal{K}_W(U)$.

The spin structure on U, together with the normal vector ∂_t, gives the product spin structure on Z. Along the initial slice, the identifications used in (3.1) are

$$\Sigma U \cong \begin{cases} (\Sigma Z)|_U, & n \text{ even}, \\ (\Sigma^\tau Z)|_U, & n \text{ odd}. \end{cases}$$

For $\psi \in \mathcal{K}_W(U)$, the spinorial Gauss formula then gives

$$\bar{\nabla}^{\Sigma Z}_X \psi = \nabla^{\Sigma U}_X \psi + \frac{1}{2} W(X) \cdot \psi = 0, \qquad X \in TU.$$

Parallel transport along the normal geodesics gives the unique candidate for an ambient parallel extension:

$$\bar{\nabla}^{\Sigma Z}_{\partial_t} \Psi = 0, \qquad \Psi|_{\{0\}\times U} = \psi.$$

The parallel spinor extension theorem of Ammann–Moroianu–Moroianu [AMM13, Theorem 3.2] shows that this extension is parallel in every direction. Here the initial slice is oriented and real analytic, and $(Z, \bar{g})$ is Ricci-flat and spin, as required. The theorem also requires $i_* : \pi_1(U) \to \pi_1(Z)$ to be surjective, where $i : U \hookrightarrow Z$ is the inclusion. In our product neighbourhood i_* is an isomorphism: Z deformation retracts onto U. Every $\psi \in \mathcal{K}_W(U)$ therefore extends to a parallel spinor on the same Z. The extension is linear, as it is defined by parallel transport. For odd n, the ambient spin connection preserves $\Sigma^\tau Z$, so the extension lies in this chosen half-spin bundle. By the spinorial Gauss formula (3.1), the restriction of any parallel spinor in the chosen ambient bundle belongs to $\mathcal{K}_W(U)$. Restriction therefore gives

$$\begin{aligned} \mathcal{P}(Z) &\xrightarrow{\cong} \mathcal{K}_W(U), \quad n \text{ even}, \\ \mathcal{P}^\tau(Z) &\xrightarrow{\cong} \mathcal{K}_W(U), \quad n \text{ odd}. \end{aligned} \tag{5.4}$$

Surjectivity follows from the extension theorem, and injectivity was established in Section 3.

5.2. **The non-flat upper bound and its attainment.** We now count the parallel spinors on the common Ricci-flat extension Z^{n+1} to bound the number of solutions for the fixed field W. Here n is the hypersurface dimension, $N = 2^{\lfloor n/2 \rfloor}$ is the intrinsic spinor rank, and d denotes the dimension of an individual factor. On a sufficiently small simply connected domain, evaluation at $p \in Z$ identifies parallel spinors with the vectors fixed by the lifted holonomy:

$$\mathcal{P}(Z) \xrightarrow[\Psi \mapsto \Psi(p)]{\cong} \left\{ \xi \in \Sigma_p Z : h \cdot \xi = \xi \text{ for all } h \in \widetilde{\mathrm{Hol}}_p(Z) \right\}.$$

Here $\widetilde{\mathrm{Hol}}_p(Z)$ is the holonomy of the spin connection. A fixed vector extends by parallel transport independently of the chosen path, and every parallel spinor is determined by its value at p. When n is odd, Z^{n+1} has even dimension, and the same correspondence holds in each half-spin bundle.

Since $\bar{g}$ is real analytic, its restricted holonomy is unchanged on connected neighbourhoods of p [KN63, Chapter II, Theorem 10.8]. After shrinking around p, the local de Rham decomposition gives

$$Z \cong \mathbb{R}^\ell \times Z_1 \times \cdots \times Z_k,$$

where the Z_i are non-flat irreducible factors. Each factor is Ricci-flat, since the Ricci tensor splits under products, and each has parallel spinors by the product rule below. A Ricci-flat locally symmetric Riemannian space is flat, so the non-flat factors are not locally symmetric. The classification of irreducible restricted holonomy applies locally to these factors. Here

Hol^0 denotes the identity component of the Riemannian holonomy group. On our simply connected domains, this is the full holonomy group. Parallel spinors correspond to vectors fixed by its lift to the spin group. For each factor Y, write $d = \dim Y$ and $p(Y) = \dim_{\mathbb{C}} \mathcal{P}(Y)$ for its number of parallel spinors. In even dimensions, write $p^{\pm}(Y) = \dim_{\mathbb{C}} \mathcal{P}^{\pm}(Y)$ for the number of each chirality; reversing orientation may interchange these counts. The table below gives Wang's counts [Wan89], with the holonomy lifts discussed in [MS00]. We take $r \geq 2$ for $\mathrm{SU}(r)$ and $r \geq 1$ for $\mathrm{Sp}(r)$; the case $\mathrm{SU}(2) = \mathrm{Sp}(1)$ appears in both families.

(5.5)

Hol^0	d	p^+	p^-	p
$\mathrm{SU}(r)$, r even	$2r$	2	0	2
$\mathrm{SU}(r)$, r odd	$2r$	1	1	2
$\mathrm{Sp}(r)$	$4r$	$r+1$	0	$r+1$
G_2	7	–	–	1
$\mathrm{Spin}(7)$	8	1	0	1

For a factor Y of dimension d, define its *full-spin efficiency* by

$$\eta(Y) := \frac{p(Y)}{2^{\lfloor d/2 \rfloor}}. \tag{5.6}$$

This is the proportion of the full spinor fibre represented by parallel spinors, denoted by κ in Alekseevsky–Cortés [AC09, Section 3]. From (5.5), the efficiencies of the non-flat irreducible factors are

$$\eta_{\mathrm{SU}(r)} = \frac{2}{2^r}, \qquad \eta_{\mathrm{Sp}(r)} = \frac{r+1}{2^{2r}}, \qquad \eta_{G_2} = \frac{1}{8}, \qquad \eta_{\mathrm{Spin}(7)} = \frac{1}{16}.$$

All are at most $1/2$, with equality precisely in the four-dimensional $\mathrm{Sp}(1)$ case. A flat factor has efficiency 1.

The full-spin efficiency is multiplicative under products [AC09, Corollary 3]. For factors Y_1, Y_2 of dimensions d_1, d_2, the full spin representation restricts to the product of the spin groups as

$$\Delta_{d_1+d_2} \cong \begin{cases} \Delta_{d_1} \otimes \Delta_{d_2}, & \text{at least one } d_i \text{ is even,} \\ (\Delta_{d_1} \otimes \Delta_{d_2})^{\oplus 2}, & \text{both } d_i \text{ are odd.} \end{cases}$$

The two holonomy groups act on their respective tensor factors. Thus, writing G_i for the lifted holonomy of Y_i,

$$(\Delta_{d_1} \otimes \Delta_{d_2})^{G_1 \times G_2} = \Delta_{d_1}^{G_1} \otimes \Delta_{d_2}^{G_2}.$$

The corresponding rule for parallel spinors is

$$\mathcal{P}(Y_1 \times Y_2) \cong \begin{cases} \mathcal{P}(Y_1) \otimes \mathcal{P}(Y_2), & \text{at least one } d_i \text{ is even,} \\ (\mathcal{P}(Y_1) \otimes \mathcal{P}(Y_2))^{\oplus 2}, & \text{both } d_i \text{ are odd.} \end{cases}$$

The number of parallel spinors and the rank of the full spinor bundle therefore acquire the same factor c, where $c = 2$ if both dimensions are odd and $c = 1$ otherwise. Consequently,

$$\eta(Y_1 \times Y_2) = \frac{c\, p(Y_1) p(Y_2)}{c\, 2^{\lfloor d_1/2 \rfloor} 2^{\lfloor d_2/2 \rfloor}} = \eta(Y_1)\eta(Y_2).$$

For a non-flat germ, the de Rham decomposition has $k \geq 1$, so $\eta(Z) = \prod_{i=1}^{k} \eta(Z_i) \leq 1/2$. This gives the full-spin bound

$$p(Z) \leq \frac{1}{2}\, 2^{\lfloor (n+1)/2 \rfloor}. \tag{5.7}$$

When n is even, Z^{n+1} has odd dimension and its full spinor bundle has rank $2^{n/2} = N$, equal to the intrinsic spinor rank. Thus (5.4) already gives $\dim_{\mathbb{C}} \mathcal{K}_W(U) = p(Z) \leq N/2$. When n is odd, the correspondence instead uses one half-spin space $\mathcal{P}^{\tau}(Z)$, and

$$\operatorname{rank}_{\mathbb{C}} \Sigma Z = 2^{(n+1)/2} = 2N, \qquad \operatorname{rank}_{\mathbb{C}} \Sigma^{\pm} Z = 2^{(n-1)/2} = N.$$

The bound (5.7) only gives $p^{+}(Z) + p^{-}(Z) \leq N$, whereas we need $p^{\tau}(Z) \leq N/2$. We therefore need a bound for each chirality separately. For two even-dimensional factors Y_1, Y_2, the combinations $(+,+)$ and $(-,-)$ give positive chirality on the product; $(+,-)$ and $(-,+)$ give negative chirality. Writing $p_i^{\pm} = p^{\pm}(Y_i)$, the product rules are

$$\begin{aligned} p^{+}(Y_1 \times Y_2) &= p_1^{+} p_2^{+} + p_1^{-} p_2^{-}, \\ p^{-}(Y_1 \times Y_2) &= p_1^{+} p_2^{-} + p_1^{-} p_2^{+}. \end{aligned} \tag{5.8}$$

In particular, if $p_1^{+} = p_1^{-} = a$, both product counts equal $a(p_2^{+} + p_2^{-})$. Thus, if either factor has equally many parallel spinors of each chirality, so does the product. If both factors are odd-dimensional, multiplication by either factor's volume form takes positive parallel spinors on the product to negative ones, and conversely. It preserves parallelness because the volume form is parallel. Its Clifford action is invertible, so the two spaces have equal dimension. The full-spin product rule gives a total of $2p(Y_1)p(Y_2)$ parallel spinors, so

$$p^{+}(Y_1 \times Y_2) = p^{-}(Y_1 \times Y_2) = p(Y_1)p(Y_2).$$

We want to bound each chiral count by one quarter of the full spinor rank. If the two counts are equal, the preceding bound on their sum immediately gives this. Otherwise, we will show that the total number of parallel spinors is itself at most one quarter of the full spinor rank.

Proposition 5.2 (Half-spin upper bound)**.** *Let $n \geq 5$ be odd. For a non-flat Ricci-flat germ Z^{n+1} with non-zero parallel spinors,*

$$\max\{p^{+}(Z), p^{-}(Z)\} \leq 2^{(n-3)/2} = N/2. \tag{5.9}$$

Proof. Since $\dim Z = n+1$ is even, the local product decomposition has an even number of odd-dimensional factors, including the flat factor when its dimension is odd. Pair these factors and keep the even-dimensional factors separately. Suppose first that one factor or pair has equal numbers of parallel spinors of each chirality. By the product rules, the same is true of Z, and (5.7) gives

$$p^{\pm}(Z) = \frac{1}{2} p(Z) \leq \frac{1}{4}\, 2^{(n+1)/2} = 2^{(n-3)/2} = N/2.$$

This case includes any flat factor of positive even dimension, any $\mathrm{SU}(r)$ factor with r odd, and any pair of odd-dimensional factors.

Suppose now that no factor or pair has equal chiral counts. By (5.5), all the factors are then even-dimensional and non-flat, with parallel spinors of only one chirality. If there are at least two factors, multiplicativity gives $\eta(Z) \leq (1/2)^2 = 1/4$. If there is only one factor, the four-dimensional $\mathrm{Sp}(1)$ case is excluded since $\dim Z = n+1 \geq 6$. Every remaining entry of

(5.5) with parallel spinors of only one chirality has full-spin efficiency at most $1/4$. In either case,

$$p^{\pm}(Z) \leq p(Z) = \eta(Z)\, 2^{(n+1)/2} \leq 2^{(n-3)/2} = N/2. \qquad \square$$

These bounds are attained by taking $Z^{n+1} = X^4 \times \mathbb{R}^{n-3}$, where X is the non-flat hyperkähler factor of Section 3.1, oriented so that $p^+(X) = 2$ and $p^-(X) = 0$. For even $n \geq 4$, the equality $\eta(Z) = \eta(X) = 1/2$ gives $p(Z) = N/2$, attaining (5.7). For odd $n \geq 5$, the flat factor has positive even dimension, with $p^{\pm}(\mathbb{R}^{n-3}) = 2^{(n-5)/2}$, so (5.8) gives

$$p^+(Z) = p^-(Z) = 2 \cdot 2^{(n-5)/2} = 2^{(n-3)/2} = N/2, \tag{5.10}$$

attaining (5.9). For $n = 7$, there is also an equality example without a flat factor: if X_1^4, X_2^4 are non-flat hyperkähler factors oriented as above, then $p^+(X_1 \times X_2) = 4$ and $p^-(X_1 \times X_2) = 0$.

For $n = 3$, however, we may take $Z = X^4$, with $p^+(Z) = 2 = \operatorname{rank}_{\mathbb{C}} \Sigma^+ Z = N$. The entire positive half-spin fibre is realised by parallel spinors despite the metric being non-flat. This explains why maximal generalised Killing spinor multiplicity on a three-dimensional hypersurface does not force the ambient metric to be flat.

Completion of the proof of Theorem D. The simultaneous extension and the correspondence were proved in (5.4). For a non-flat germ, if n is even, (5.7) gives $\dim_{\mathbb{C}} \mathcal{K}_W(U) = p(Z) \leq 2^{n/2-1} = N/2$. If n is odd and $n \geq 5$, Proposition 5.2 gives $\dim_{\mathbb{C}} \mathcal{K}_W(U) = p^\tau(Z) \leq 2^{(n-3)/2} = N/2$. The equality examples above and the hypersurfaces of Section 3.1 show that the bound is attained for every $n \geq 4$. $\square$

Corollary 5.3 (Forcing for analytic data)**.** *Let $n \geq 4$ and suppose that (g, W) is real analytic. If $\dim_{\mathbb{C}} \mathcal{K}_W(U) > N/2$ on a sufficiently small connected, simply connected domain $U \subset M$, then the common Ricci-flat germ is flat. Hence (g, W) satisfies the Euclidean Gauss–Codazzi equations on U.*

Proof. If the germ were non-flat, its number of parallel spinors would contradict the upper bound in Theorem D. Applying the Gauss and Codazzi formulae to the hypersurface in the flat germ gives the conclusion. $\square$

Appendix A. The parity-restricted zero-weight estimate

Here we prove Theorem 4.2. The problem is to estimate, for $\mu \in \mathbb{R}^q \setminus \{0\}$, the number of $\varepsilon \in \{\pm 1\}^q$ satisfying simultaneously

$$\sum_{j=1}^{q} \varepsilon_j \mu_j = 0, \qquad \prod_{j=1}^{q} \varepsilon_j = \sigma,$$

and to determine the coefficients when this number is exceptionally large.

A.1. **The antichain estimate.** A family $\mathcal{A}$ of subsets of $[q] = \{1, \ldots, q\}$ is an antichain if no two distinct members are related by inclusion. The Lubell–Yamamoto–Meshalkin inequality [Lub66, Yam54, Mes63] gives

$$\sum_{A \in \mathcal{A}} \binom{q}{|A|}^{-1} \leq 1. \tag{A.1}$$

For the proof we refer to the works cited. Since each binomial coefficient is at most the central binomial coefficient, Sperner's bound [Spe28]

$$|\mathcal{A}| \le \binom{q}{\lfloor q/2 \rfloor} \tag{A.2}$$

follows immediately. Indeed, the left-hand side of (A.1) is at least $|\mathcal{A}|/\binom{q}{\lfloor q/2 \rfloor}$.

A.2. **Changing signs and zero coefficients.** For each j write $\mu_j = s_j a_j$, with $a_j = |\mu_j|$ and $s_j \in \{\pm 1\}$; if $\mu_j = 0$, set $s_j = 1$. Setting $\varepsilon'_j = s_j \varepsilon_j$,

$$\sum_j \varepsilon_j \mu_j = \sum_j \varepsilon'_j a_j, \qquad \prod_j \varepsilon'_j = \sigma \prod_j s_j. \tag{A.3}$$

This is a bijection of sign vectors, so the signs of the coefficients can be removed at the cost, if necessary, of interchanging the two parity classes.

We first treat the case in which some coefficient vanishes. After relabelling we may assume $\mu_1 \neq 0$ and $\mu_q = 0$. Once $\varepsilon_2, \dots, \varepsilon_{q-1}$ are fixed, the zero-sum condition determines ε_1 in at most one way. Once it is determined, the parity condition determines ε_q uniquely. Hence

$$|Z^\sigma(\mu)| \le 2^{q-2} \qquad \text{(when some coefficient is zero)}. \tag{A.4}$$

This case has to be treated separately, because the antichain property of positive weights used later may fail when there are zero coefficients.

A.3. **The sharp zero-weight upper bound.** Suppose that all coefficients are non-zero. Set $a_j = |\mu_j| > 0$ and $\sigma' = \sigma \prod_j s_j$. The sign change (A.3) gives $|Z^\sigma(\mu)| = |Z^{\sigma'}(a)|$. Set $T = \sum_j a_j$. Associating to a sign vector ε' its set of positive indices $A = \{j : \varepsilon'_j = 1\}$, we have

$$\sum_j \varepsilon'_j a_j = 0 \iff \sum_{j \in A} a_j = T/2, \qquad \prod_j \varepsilon'_j = (-1)^{q-|A|}. \tag{A.5}$$

Hence prescribing the parity class prescribes the parity of $|A|$. Let $\mathcal{F}$ be the family of all subsets of weight $T/2$. If $A \subsetneq B$, positivity gives $\sum_{j\in A} a_j < \sum_{j \in B} a_j$, so $\mathcal{F}$ is an antichain. Moreover, if $A \in \mathcal{F}$ then its complement A^c also belongs to $\mathcal{F}$.

When q is odd, complementation interchanges the parity of the cardinality. Hence the numbers of zero weights in the two parity classes are equal, and (A.2) gives

$$|Z^{\sigma'}(a)| = \frac{1}{2}|\mathcal{F}| \le \frac{1}{2}\binom{q}{(q-1)/2} \le 2^{q-2}.$$

The last inequality holds because, for odd q, there are two equal largest binomial coefficients, whose sum is at most 2^q.

When q is even, the subfamily $\mathcal{A} \subset \mathcal{F}$ of the chosen parity is itself an antichain, so

$$|Z^{\sigma'}(a)| = |\mathcal{A}| \le \binom{q}{q/2}.$$

Combining this with (A.4) gives (4.3).

We check sharpness. For $q \ge 3$ take $\mu = (1, 1, 0, \dots, 0)$. The zero-sum condition is $\varepsilon_1 = -\varepsilon_2$. Since the product of the remaining $q-2$ signs is prescribed, each parity class contains $2 \cdot 2^{q-3} = 2^{q-2}$ solutions. On the other hand, for even q take $\mu = (1, \dots, 1)$. The zero weights are then exactly those with $q/2$ positive and $q/2$ negative signs, and there are $\binom{q}{q/2}$ of them.

They all belong to the same parity class. Changing the sign of one coefficient interchanges the two parity classes. Sharpness for $q = 2$ is also given by the latter example. Hence the upper bound is attained for each fixed σ.

We determine for which even q the central binomial coefficient exceeds half rank.

q	2	4	6	8	10
$\binom{q}{q/2}$	2	6	20	70	252
2^{q-2}	1	4	16	64	256

Setting $c_q = \binom{q}{q/2}/2^{q-2}$, we have

$$\frac{c_{q+2}}{c_q} = \frac{q+1}{q+2} < 1,$$

so the exceptions are exactly $q = 2, 4, 6, 8$.

A.4. **Rigidity in the exceptional cases.** We first prove a lemma forcing the coefficients to have equal size.

Lemma A.1 (Middle-layer swap lemma)**.** *Let $q = 2m$, $a_j > 0$ and $T = \sum_j a_j$. Suppose that each member of a family $\mathcal{A}$ of m-element subsets has weight $T/2$. If fewer than $\binom{q-2}{m-1}$ of all m-element subsets are missing from $\mathcal{A}$, then $a_1 = \cdots = a_q$.*

Proof. Fix distinct i, j, and for $B \subset [q] \setminus \{i, j\}$ with $|B| = m - 1$, pair

$$\{i\} \cup B, \qquad \{j\} \cup B.$$

There are $\binom{q-2}{m-1}$ such pairs, and they are pairwise disjoint. Fewer than $\binom{q-2}{m-1}$ subsets are missing, so at least one pair has both members in $\mathcal{A}$. Both have weight $T/2$ and share the same subset B. Cancelling the weights of the elements of B gives $a_i = a_j$. This holds for every pair $i \neq j$, so $a_1 = \cdots = a_q$. □

Assume $q \in \{4, 6, 8\}$ and $|Z^\sigma(\mu)| > 2^{q-2}$. By (A.4) all coefficients are non-zero, and after changing signs we may assume they are all positive. Let $\mathcal{A}$ be the family of zero-weight subsets of the prescribed parity, and set

$$b_k := |\{A \in \mathcal{A} : |A| = k\}|.$$

Since q is even, complementation preserves this family, and $b_k = b_{q-k}$. Moreover, complementation has no fixed points, so $|\mathcal{A}|$ is even. The empty set and the whole set do not have weight $T/2$. If a singleton $\{i\}$ belongs to the family, so does its complement. Any other member either contains i, and so properly contains $\{i\}$, or does not contain i, and so is properly contained in the complement. By the antichain property, in this case $|\mathcal{A}| = 2$.

The case $q = 4$

If the chosen parity is odd, the possible sizes are 1 and 3, and the LYM inequality gives

$$\frac{b_1}{4} + \frac{b_3}{4} \leq 1, \qquad |\mathcal{A}| \leq 4,$$

contrary to the hypothesis. Hence every member has size 2. There are 6 such subsets in total, and since $|\mathcal{A}|$ is even and greater than 4, we get $|\mathcal{A}| = 6$. No middle-layer subsets are missing, and since $0 < \binom{2}{1} = 2$, Lemma A.1 applies.

The case $q = 6$

If the chosen parity is even, the possible sizes are 2 and 4, and

$$\frac{b_2}{15} + \frac{b_4}{15} \leq 1$$

gives $|\mathcal{A}| \leq 15 < 16$. Hence the parity is odd, and the sizes are 1, 3 and 5. The observation on singletons excludes sizes 1 and 5, so every member has size 3. Since $\binom{6}{3} = 20$, $|\mathcal{A}| > 16$, and $|\mathcal{A}|$ is even, at most two subsets are missing. Since $2 < \binom{4}{2} = 6$, Lemma A.1 applies.

The case $q = 8$

If the odd parity is chosen and a singleton is a member, there are two members. Otherwise only the sizes 3 and 5 are possible, and

$$\frac{b_3}{56} + \frac{b_5}{56} \leq 1$$

gives $|\mathcal{A}| \leq 56 < 64$. Hence the parity is even, and the sizes are 2, 4 and 6. From $b_2 = b_6$ and the LYM inequality,

$$\frac{2b_2}{28} + \frac{b_4}{70} \leq 1, \qquad b_4 \leq 70 - 5b_2, \qquad |\mathcal{A}| \leq 70 - 3b_2.$$

If $b_2 \geq 2$, the count is at most 64. Suppose $b_2 = 1$, and let P be the unique two-element member of $\mathcal{A}$. Its complement P^c is the unique six-element member. A four-element member A cannot contain P, since $\mathcal{A}$ is an antichain. Nor can A be disjoint from P: then $A \subset P^c$, again contradicting the antichain property. Thus $|A \cap P| = 1$. There are at most $\binom{2}{1}\binom{6}{3} = 40$ such subsets, so

$$|\mathcal{A}| \leq 1 + 40 + 1 = 42 < 64.$$

Hence $b_2 = b_6 = 0$, and every member has size 4. Since $\binom{8}{4} = 70$ and $|\mathcal{A}|$ is even, $|\mathcal{A}| \geq 66$, so at most four subsets are missing. Since $4 < \binom{6}{3} = 20$, Lemma A.1 applies.

In all three cases we have obtained $a_1 = \cdots = a_q$. Restoring the original signs gives $|\mu_1| = \cdots = |\mu_q| \neq 0$. After the sign change, the zero-sum condition is equivalent to there being exactly $q/2$ positive signs. Hence there are $\binom{q}{q/2}$ zero weights, all in the same parity class, and none in the opposite class. This completes the proof of Theorem 4.2.

AI use

Large language models were used extensively during the research and preparation of this article to explore proof strategies, derive and check formulas, examine arguments for gaps, and assist with exposition. In particular, they contributed substantially to developing the arguments for the exceptional dimensions 7, 11 and 15 in the proof of the smooth threshold theorem. The authors take full responsibility for the content and correctness of the article.

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